PANTONE REPORT : THE PAGES OF MICHEL DAVID

PAGE 14

OPERATION OF THE PANTONE REACTOR WITH AIR BLOCKING DEVICE ALLOWING TO RUN A GASOLINE ENGINE ON DOMESTIC FUEL

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In my pages 5 and 10 I have described a successfully tested air shutter device.

This device makes it possible to put an end to the vacuum problems necessary to make the Pantone reactor operate correctly, especially when the latter is supplied with fatty fuel type fuel oil by a spray device of the carburetor type.

Indeed, as I have often repeated in my pages, it is essential that everything that passes in the reactor be done with sufficient speed and this at all engine speeds (see page 13).

However, in many assemblies, the air intake between the reactor outlet and the engine inlet inlet causes insoluble problems in the operation of reactors provided with carburettors. This is normal since the opening of this air intake favors the entry of air to the detriment of the vacuum in the reactor.

I had therefore proposed in my pages 5 and 10 a rotary valve system which closes the air inlet during part of the admission time, so that during this closure, only the reactor is requested by the suction the engine, which allows operation with the carburetor in the choke position.

It should be noted that it was only thanks to this air valve device that I managed to make gasoline engines work under good conditions with heating oil, pure alcohol or mixed with up to 40% water.

Although of simple design, the air intake shutter valve described on my page 10 has the disadvantage of not being within the reach of those who do not have machine tools. There is also the disadvantage that a large part of the small engines equipping most generator sets or other devices do not offer access to the motor shaft to install a pinion for the toothed belt of this rotary valve.

For my small 2 kW Honda generator, I therefore imagined a device that did not have to be connected to the motor shaft.

The device consists of a perforated disk rotating in front of the air inlet slot and driven by a small electric motor taken from a cassette tape recorder or other small electric motor.

It is therefore a real assembly that can be done without machine tools, but of formidable efficiency.

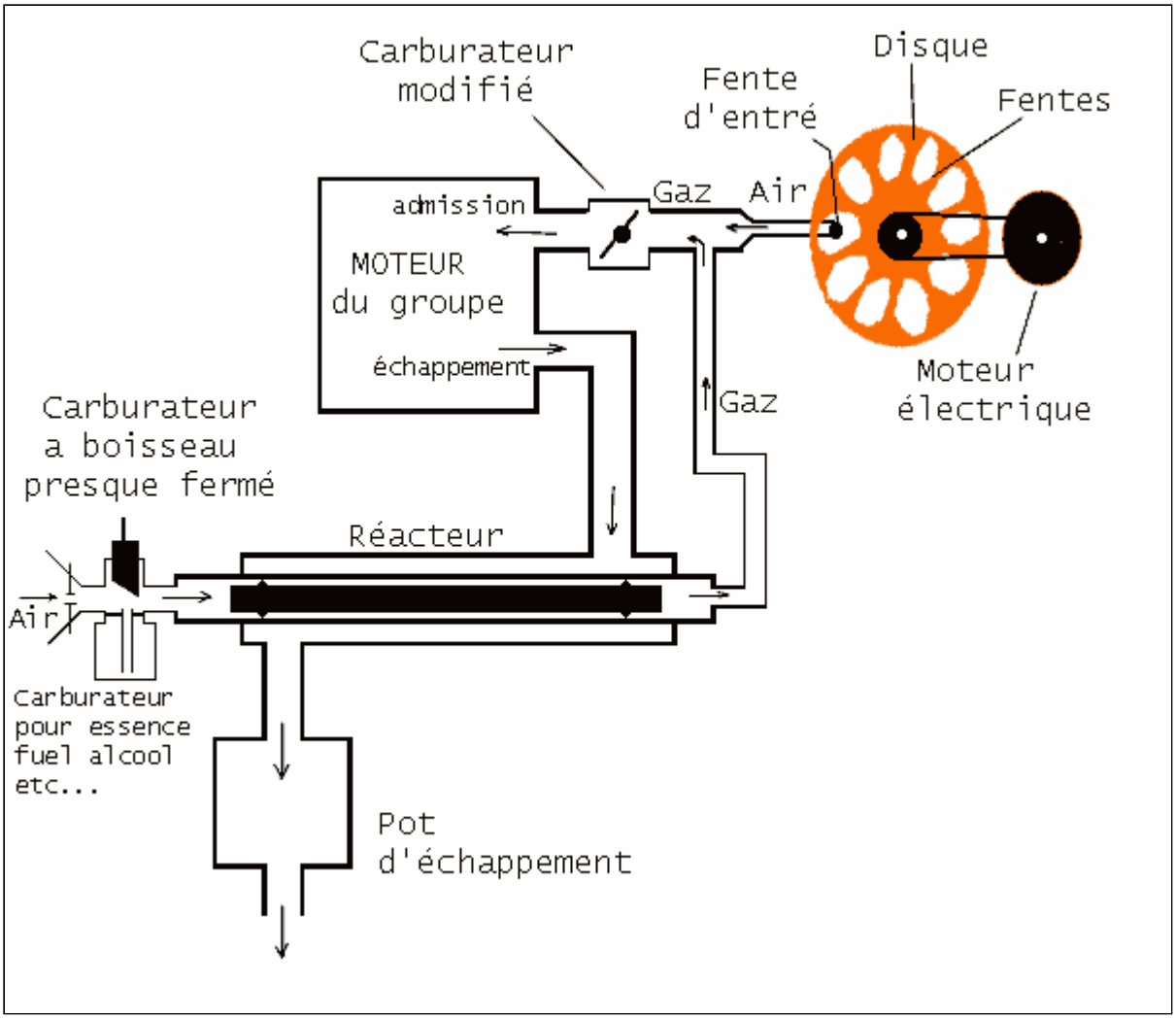
This assembly operates at constant speed independently of the speed of the heat engine.

The speed of rotation of the disc is around 4000 rpm in my case. Taking into account the number of openings on the disc there are several openings and closings of the air intake during the intake time even when the engine is running at 3000 rpm which is the normal speed of this small generator.

For the sake of simplicity, I used the ball bearings and the disc of a disused computer hard drive, the rest is made of a PVC plate and resin paste to be mixed by hand which allows to make the collages and shapes that you want.

See photos below.

The important advantage is also that the engine speed is only controlled by a single valve which can be coupled to the generator set control lever.



If you manage to build this type of air shutter or something similar, you will achieve surprising results and put an end to many of the problems you have with the Pantone reactor: it will take you a significant leap in your work.

Note: It is obvious that most of the devices that I present in these pages are solutions likely to be implemented by more or less well-equipped handymen.

These devices make it possible to operate Pantone reactors without having to modify the mechanics - engine.

One suspects that it would be possible to obtain better results in many cases if the engines were designed differently to use the Pantone reactor.

One could, for example, to avoid the air shut-off valve, provide an additional intake valve especially for the reactor, and an air compressor to help fill the engine with air. In short, from the moment we master the operation of the reactor, the door is open to all kinds of possibilities.

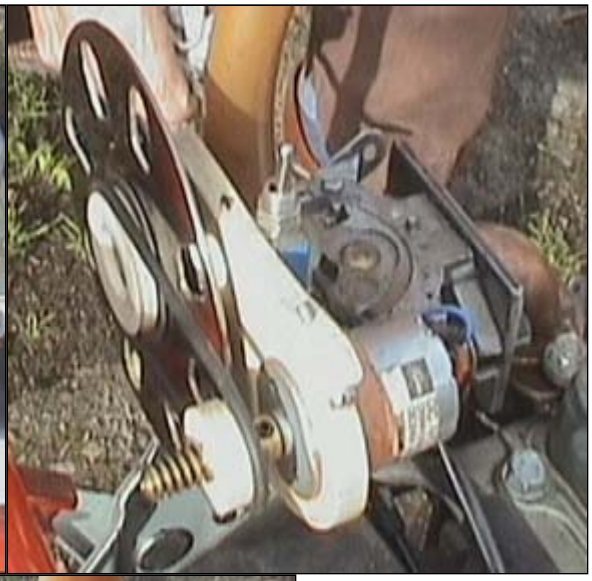
General view of the small Honda group of 2.2 kW

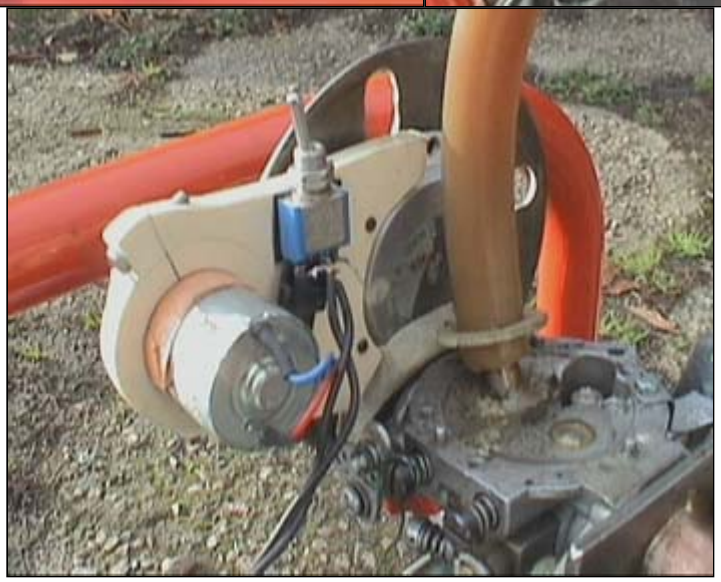
Top view: we can clearly see the engine, the carburetor and the exhaust.
This reactor consists of a rod 8 mm in diameter, 16 cm in length in a 10 mm inner tube.

Here we distinguish the modified carburettor of which we only use the butterfly.
The reactor suction inlet and the air inlet slit which are on the same side of the throttle (see diagram).
Note: plug all the small nozzle holes and others: no parasitic air intakes.

View of the carburetor inlet in choke position

Three views of the small electric motor driving the shutter disc. Note the pulleys and the rubber belt.



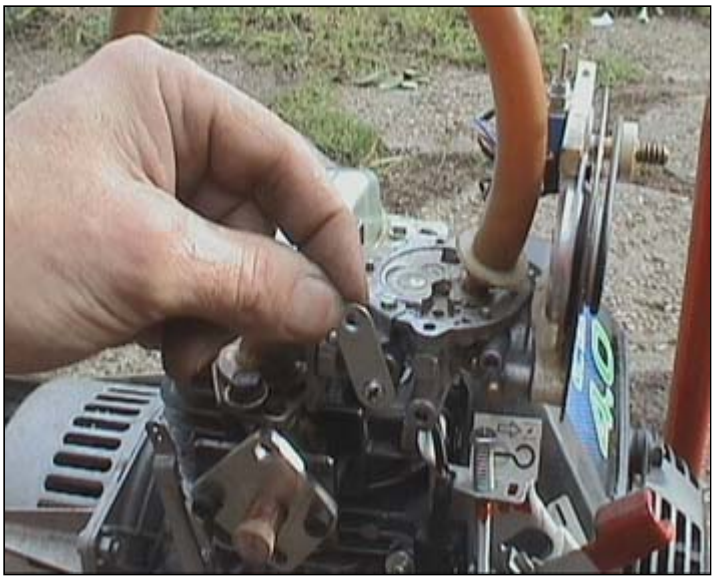


The motor can be powered by batteries or mains power etc.

One of the disc openings can be seen in front of the air inlet slot.

Here the air inlet slit is closed.
The air inlet slot is cut in the PVC plate screwed in front of the modified carburetor inlet, in place of the air filter. This carburetor only serving as a place for mixing Pantone gas and air as well as an acceleration valve

Below is the flexible silicone hose, which connects the outlet of the reactor to the modified carburetor of which <u>only the throttle is used</u> to control engine speed.



Here we see the lever attached to the throttle shaft of the modified carburetor which serves as the engine acceleration valve

Mr. David and Xavier Faure for the photos and the animated diagram

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